question 5 theory part

Convergence of ADALINE to the Global Least-Squares Solution

The ADALINE (Adaptive Linear Neuron) model proposed by Bernard Widrow and Ted Hoff learns by minimizing the Mean Squared Error (MSE) between the predicted output and the desired output. Unlike the perceptron, ADALINE uses a linear activation during training, which allows it to apply gradient descent on a quadratic error surface. Because the error surface is convex, ADALINE has a strong convergence guarantee to the global least-squares solution.

1. ADALINE Error Function

For an input vector x with weight vector w and desired output d:

$$y = w^T x$$

Error:

$$e = d - y$$

For N training samples, the Mean Squared Error (MSE) cost function is:

$$E(w) = \frac{1}{2} \sum_{i=1}^{N} (d_i - w^T x_i)^2$$

This can also be written in matrix form:

$$E(w) = \frac{1}{2} \|Xw - d\|^2$$

where

- X = input matrix
- w = weight vector
- d = desired output vector

2. Least-Squares Solution

To minimize the error, differentiate $E(w)$ with respect to w:

$$\nabla E(w) = -X^T(d - Xw)$$

Setting the gradient to zero:

$$X^T(d - Xw) = 0$$

$$X^T Xw = X^T d$$

Solving for w:

$$w^* = (X^T X)^{-1} X^T d$$

This solution w^* is the least-squares optimal weight vector.

Since the cost function is quadratic and convex, it has only one minimum, therefore:

$$w^* = \text{global minimum}$$

Thus ADALINE converges to the global least-squares solution.

3. Gradient Descent Learning Rule

ADALINE updates weights using the Widrow-Hoff (Delta) rule:

$$w_{new} = w_{old} + \eta(d - y)x$$

where

- η = learning rate
- $(d - y)$ = error

This rule performs gradient descent on the quadratic error function.

Because the error surface is a convex paraboloid, gradient descent will always converge to the unique global minimum if the learning rate is sufficiently small.

4. Strong Convergence Guarantee

The convergence of ADALINE can be shown through the properties of its error function.

Quadratic Cost Function

$$E(w) = \frac{1}{2}(d - w^T x)^2$$

is quadratic in w.

Convex Error Surface

The Hessian matrix is:

$$H = X^T X$$

Since $X^T X$ is positive semi-definite, the error surface is convex.

Unique Minimum

A convex quadratic function has one unique global minimum, ensuring that gradient descent converges to that solution.

Convergence Condition

ADALINE converges when the learning rate satisfies:

$$0 < \eta < \frac{2}{\lambda_{max}}$$

where λ_{max} is the largest eigenvalue of $X^T X$.

Under this condition, weight updates monotonically decrease the error until reaching the least-squares solution.

5. Independence from Data Separability

Unlike the perceptron, ADALINE does not require linearly separable data.

Reason

Perceptron learning relies on a hard threshold function:

$$y = \mathrm{sign}(w^T x)$$

If the dataset is not linearly separable, perceptron learning fails to converge.

However, ADALINE minimizes the continuous squared error function:

$$E(w) = \frac{1}{2} \sum (d_i - w^T x_i)^2$$

Even if perfect classification is impossible:

- ADALINE still finds the best approximate solution
- It chooses the weight vector that minimizes overall squared error

Thus, ADALINE always converges to the least-squares solution, regardless of separability.

6. Key Result

ADALINE guarantees convergence because:

- The error surface is convex and quadratic
- Gradient descent finds the unique global minimum
- The solution corresponds to the least-squares estimator

Therefore:

ADALINE converges to the global least-squares solution and its convergence does not depend on whether the training data is linearly separable.